1. Introduction to ROS 2

1. Overview of ROS 2

ROS 2 is the second-generation Robot Operating System, an upgraded version of ROS1 that solves some of the problems existing in ROS1. The earliest ROS 2 version, Arden, appeared in 2017. With version iteration, continuous updating, and optimization, a stable version is now available. As with ROS1, the choice of the Linux version is also related to the ROS 2 version, and the corresponding versions are as follows.

ROS 2 Version	Release Date	EOL Date	Ubuntu Version
Ardent Apalone	2017.12	2018.12	Ubuntu 16.04 (Xenial Xerus)
Bouncy Bolson	2018.7	2019.7	Ubuntu 16.04 (Xenial Xerus), Ubuntu 18.04 (Bionic Beaver)
Crystal Clemmys	2018.12	2019.12	Ubuntu 18.04 (Bionic Beaver)
Dashing Diademata	2019.5	2021.5	Ubuntu 18.04 (Bionic Beaver)
Eloquent Elusor	2019.11	2020.11	Ubuntu 18.04 (Bionic Beaver)
Foxy Fitzroy	2020.6	2023.5	Ubuntu 20.04 (Focal Fossa)
Galactic Geochelone	2021.5	2022.11	Ubuntu 20.04 (Focal Fossa)
Humble Hawksbill	2022.5	2027.5	Ubuntu 22.04 (Jammy Jellyfish)
Iron Irwini	2023.5	2024.11	Ubuntu 22.04 (Jammy Jellyfish)
Jazzy Jalisco	2024.5	2029.5	Ubuntu 24.04 (Noble Numbat)

- **Note**: Select the ROS 2 version corresponding to your Linux distribution. This course series targets **Ubuntu 24.04 + ROS 2 Jazzy**, and the commands and cases in the course are all adapted to the Jazzy environment. Other ROS 2 versions may differ in package names, interfaces, and launch files, and cannot be copied directly.

2. Features of ROS 2

2.1. **ROS 2 Fully Supports Three Platforms**

- Ubuntu
- Mac OS X
- Windows 10

2.2. Implemented a Distributed Architecture

The master central node is eliminated, enabling distributed node discovery, publish/subscribe, and request/response communication.

2.3. **Real-Time Support**

2.4. Uses New Versions of Programming Languages

- C++17
- Python 3.12

2.5. **Uses the New Build System Ament (Catkin for ROS)**

2.6. **ROS1 Can Communicate with ROS 2 via rosbridge**

3. Differences Between ROS 2 and ROS1

3.1. Platform

ROS1 currently only supports running on the Linux system, and is commonly built and used in Ubuntu. ROS 2, on the other hand, can be built and used in Ubuntu, Windows, and even on embedded development boards, so its platform support is broader.

3.2. Language

- C++

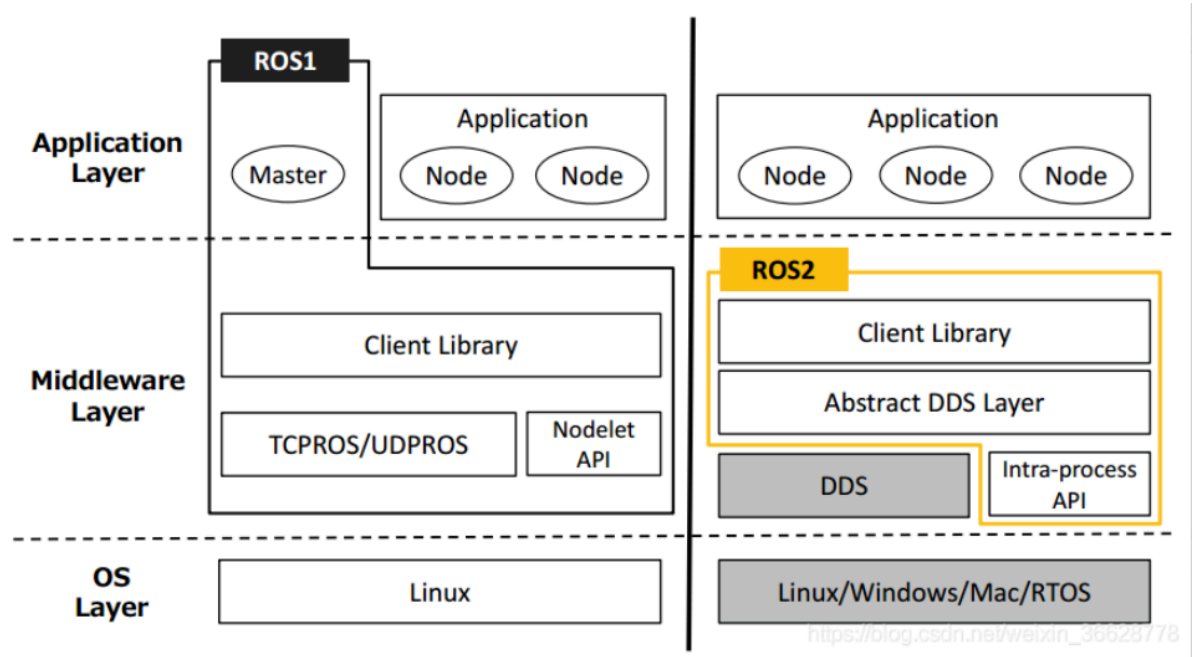
  The core of ROS1 is C++03, while ROS 2 makes extensive use of C++11.

- Python

  In its early days, ROS1 mainly used Python 2, while ROS 2 uses Python 3; ROS 2 Jazzy on Ubuntu 24.04 uses Python 3.12.

3.3. Middleware

Before ROS1 starts up, roscore must be started. This master controls all communication between nodes, while ROS 2 does not have one; it only has an abstract middleware interface, through which data is transmitted. Currently, all implementations of this interface are based on the DDS standard. This enables ROS 2 to provide various high-quality QoS service policies, thereby improving communication across different networks.



3.4. Compilation Commands

The ROS1 compilation command is catkin_make, while ROS 2 uses the colcon build command.